

M2: See how typos and added documents change search

Beyond Benchmarks: Robustness for Real-World Information Systems

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Learning objective: Measure how a typo changes the query vector. Then check how an added attack document changes the ranking.

Two changes to test

- Query Embedding Drift: add typos and measure how much the query vector changes. Keep the papers fixed.
- Document Insertion: add a document and check the ranking. Keep the query fixed.

Both tests keep the model weights fixed. M1 explains the dataset, relevance labels (qrels), and search steps.

Setup

Run this notebook on its own with Restart Kernel → Run All. Keep `resources/` next to the notebooks. See the README for setup and cache help.

The setup checks the SciFact files. The experiments read saved MiniLM vectors and rankings, including rankings after document insertion. They do not run the models again or change the supplied collection.

```
# Load the libraries, SciFact data, and saved model results.
from pathlib import Path
import sys
ROOT = next(p / "resources" for p in [Path.cwd(), *Path.cwd().parents] if (p / "resources" /
"tutorial_utils").is_dir())
if str(ROOT) not in sys.path:
    sys.path.insert(0, str(ROOT))
%matplotlib inline
from tutorial_utils import load_tutorial_context
from tutorial_utils import m2_lesson as lesson
from tutorial_utils import plotting as viz
from tutorial_utils.presentation import install_plain_display
install_plain_display()
MODE = "cached"
ctx = load_tutorial_context(mode=MODE, seed=42)
print(ctx.banner())
```

```
MODE=cached | SciFact | 5183 documents | 300 test queries
BM25 / MiniLM dense exact retrieval → MiniLM cross-encoder (20 candidates)
```

Experiment Settings

Choose one of M1's three queries. Set `severity` to the maximum character edits, from 0 to 3. Zero leaves the query unchanged. A short query may receive fewer edits than the maximum.

Then select the code cell below and use Run All Below. `severity` affects only the typo test. The insertion test always uses the original query and one added document.

```
# Choose the query and maximum number of character edits.
query_id = "1303"           # Tirasemtiv (default); "1012": radioiodine; "628": HTLV-1.
severity = 1                # typo only; maximum character edits: 0–3
```

Query Embedding Drift: How much does the vector change?

A typo can change the query vector's direction. We measure that change with 1 − cosine similarity. For original vector $e(q)$ and typo vector $e(q')$:

$$d(q, q') = 1 - \cos(e(q), e(q')).$$

Zero means the directions match. A larger value means they differ more. This distance does not measure search quality. Check the returned papers against qrels to see whether evidence was lost.

The output shows the edited words and vector distance.

```
# Show how a typo changes the selected query's vector.
query_demo = lesson.show_query_shift(ctx, query_id, severity)
```

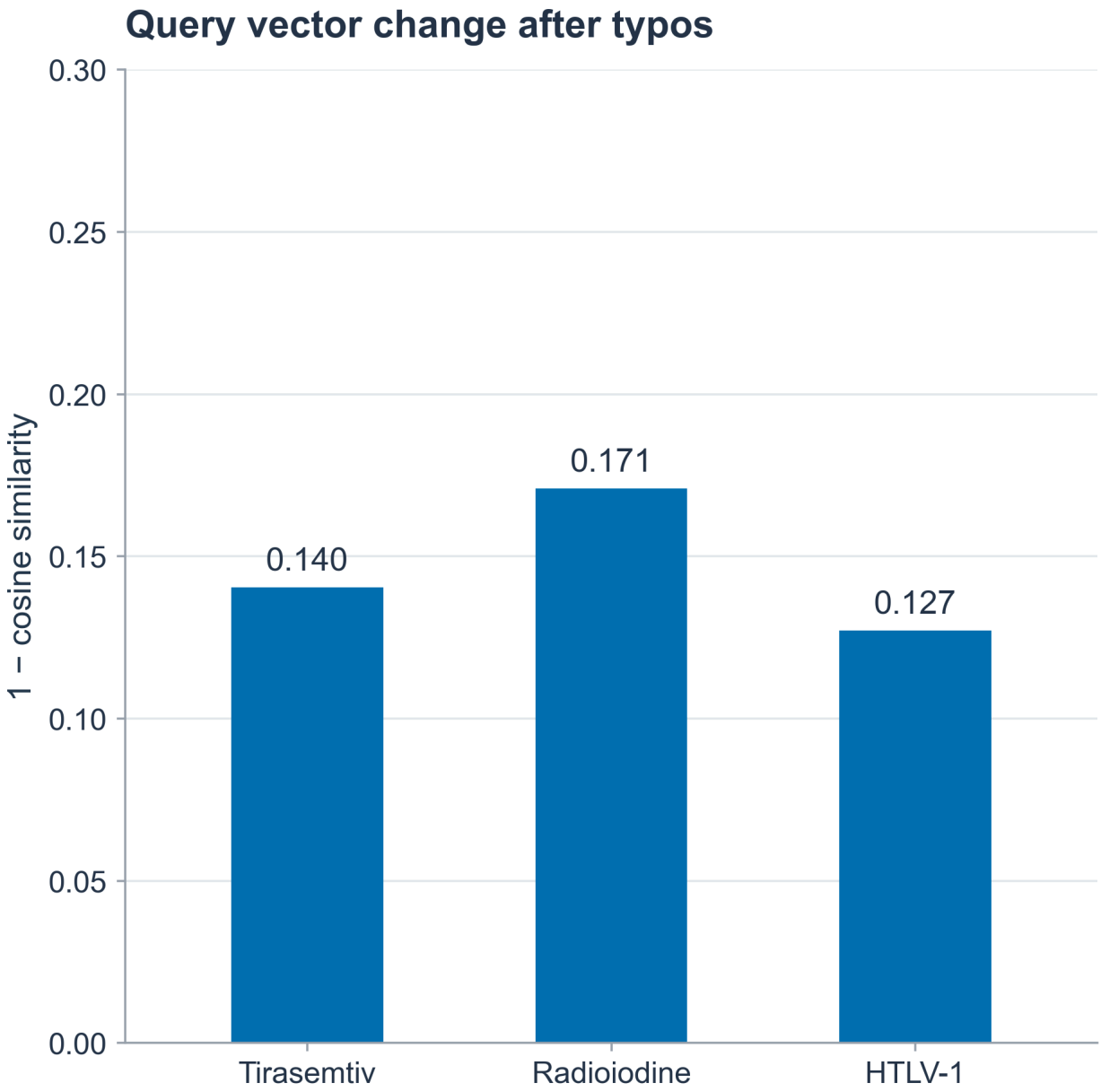
```
Selected test query, maximum character edits: 1, actual edits: 1
Original Query: Tirasemtiv has no effect on fast-twitch muscle.
Typo Query: Tirasemtiv has no effect on fast-twitch musckle.
Edited words: muscle → musckle
Embedding distance (1 − cosine similarity): 0.140
The typo changed the query vector's direction.
```

Interpretation: Even one character edit turns the query vector away from the original (distance 0.140), which can change which papers are closest to it in dense search.

Compare the three queries

Each bar shows the vector distance between an original query and its typo version. All bars use the same scale. A shorter bar means less vector change. Check relevance separately.

```
# Plot query vector changes for the three example queries.
viz.show(lesson.embedding_figure(ctx, query_id, severity))
lesson.show_embedding_interpretation(ctx, query_id, severity)
```



Interpretation: Radioiodine moves the most (0.171) yet keeps its dense nDCG@10, while Tirasemtiv moves less (0.140) and loses every known relevant paper from its dense top ten, so a longer bar does not mean a larger loss.

Document Insertion: Can a repeated claim outrank evidence?

Add one artificial passage that repeats the claim but gives no scientific evidence. Keep the query and original papers fixed. Check where the attack ranks and which papers leave the top ten.

Attack Hit@10 is 1 when the attack appears in the top ten, and 0 otherwise. For the defender, 0 is better. Rank 1 is first. `absent` means the attack is outside the 20 candidates.

`[relevant]` marks a paper with a positive qrel. A high model score alone does not make a paper relevant.

```
# Compare paper rankings before and after inserting an attack document.
insertion_demo = lesson.show_insertion(ctx, query_id)
```

Selected test query, one added document
Tirasemtiv has no effect on fast-twitch muscle.
Inserted document (cached experiment input): `Synthetic teaching passage. Tirasemtiv has no effect on fast-twitch muscle. Tirasemtiv has no effect on fast-twitch muscle. This unverified claim is intentionally repeated; this is not scientific evidence.`

Inserted document rank and attack Hit@10 (0 is safer)

Stage	Corpus	Inserted rank / 20	Hit@10
Dense	Original	absent	0
Dense	After insertion	1	1
Reranker	Original	absent	0
Reranker	After insertion	1	1

The attack ranks 1 in dense search and 1 after reranking. Still in the reranked top ten: yes.

Interpretation: A passage that only repeats the claim outranks every real paper in dense search and after reranking, so reranking does not stop this attack.

Dense top-10 (Inserted = attack; [relevant] = positive qrel; rank 1 is highest)

Rank	Original	After insertion
1	12631697 [relevant]	Inserted
2	43566999	12631697 [relevant]
3	30101891	43566999
4	34328964	30101891
5	37037012	34328964
6	23737024	37037012
7	19994006	23737024
8	33554389	19994006
9	6549091	33554389
10	23865182	6549091

Known relevant papers rank 1 before insertion and 2 after insertion. Original papers pushed out of the top ten: 1.

Interpretation: In dense search, the attack takes rank 1 and the relevant paper slips from rank 1 to 2; one original paper is pushed out of the top ten.

Reranker top-10 (Inserted = attack; [relevant] = positive qrel; rank 1 is highest)

Rank	Original	After insertion
1	12631697 [relevant]	Inserted
2	40631095	12631697 [relevant]
3	23737024	40631095
4	34328964	23737024
5	37037012	34328964
6	1275505	37037012
7	33554389	1275505
8	32322418	33554389
9	25475802	32322418
10	29526125	25475802

Known relevant papers rank 1 before insertion and 2 after insertion. Original papers pushed out of the top ten: 1.

Interpretation: Reranking keeps the attack at rank 1 and the relevant paper slips from rank 1 to 2, so an LLM would read the attack before any real paper.

Original papers pushed out of the top ten (IDs): Dense: 23865182; Reranker: 29526125

Check what happened to the evidence

Vector distance measures query change. The insertion tables show which papers moved or disappeared. To judge search quality, check the relevant papers.

Use M1's candidate-recall check: did search find the evidence, and where did reranking place it? M5 checks which papers reach the LLM.

Lesson Summary

Typos can change query vectors. An added passage can outrank real papers, even after reranking. Check which relevant papers are affected before choosing a repair.

Next: M3: Measure retrieval robustness.

Sources to read

- Zhuang & Zuccon (EMNLP 2021), Dealing with Typos for BERT-based Passage Retrieval and Ranking.
- Zhong et al. (EMNLP 2023), Poisoning Retrieval Corpora by Injecting Adversarial Passages.